

Canberra SmartWaste

Bijay Pokhrel · Samir Bhandari · Krishna Trivedi
Ayush Ale · Charanpal Kaur
Simulated TCCS / Canberra pilot — not live IoT hardware

A smart waste monitoring and citizen reporting prototype for the ACT
ICT308 Project 2 · BIT · Semester 2, 2026 · Iteration 1 completed system

PROBLEM

Fixed collection days send trucks to empty bins and still leave busy public bins overflowing. Fuel is wasted and streets get dirty.

SOLUTION

A role-based software prototype. Simulated Canberra bin fill data, a rule-based pickup score, and a citizen report form with **WST-2026-XXXX** tracking. No black-box A.I. routing.

SYSTEM OVERVIEW

Citizen — report overflow / damage, track status (Submitted → Under Review → Scheduled → Resolved).

Collection staff — only bins ≥80% full are on the run. Empty bins are skipped. Mark Collected resets fill to 5%.

Administrator — dashboard, map, alerts, assign Route ACT-R104, maintenance tickets, reset demo data.

One shared state (AppContext). Persistence is browser `localStorage` in `dataStore.ts` — there is no cloud SQL database in this delivery.

IMPLEMENTATION HIGHLIGHTS

Fill	Status
0–49%	Normal — skip truck
50–79%	Moderate — skip truck
80–89%	Collection required
90–100%	Critical — go first if score is highest

Priority score = $\text{fill} \times 0.5 + \text{urgency} + \text{overdue time}$. Collection closes only reports with the same bin ID.

GITHUB / DEMO



github.com/keinitxaminamalai-art/Smart-waste-management-system

Laptop demo: `npm install &&`
`npm run demo` → `http://localhost:5173/`

admin / staff / citizen
@smartwaste.demo

KEY TECHNOLOGIES

- React 19, TypeScript 6, Vite 8
- AppContext + `localStorage`
- Vitest, ESLint, GitHub Actions
- Lucide + Recharts

RESULTS

- Three working roles and end-to-end report → collect story
- 22 automated tests on the business rules
- Feature branches and PRs per team member
- Honest scope: simulated telemetry, not IoT

FUTURE WORK

- Live bin sensors / API
- Shared database and real login
- Geospatial shortest-path routing

ASK US

How does the driver skip empty bins? Why `localStorage`? Where is the database? We can walk the demo on a laptop in two minutes.

